



Diseases of the Immune System

Lec. 10

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Immunity

Immunity refers to protection against infections. The immune system is the collection of cells and molecules that are responsible for defending the body against the countless pathogens that individuals encounter.

Defense against pathogens consists of two types of reactions :

1. Innate immunity
2. Adaptive immunity



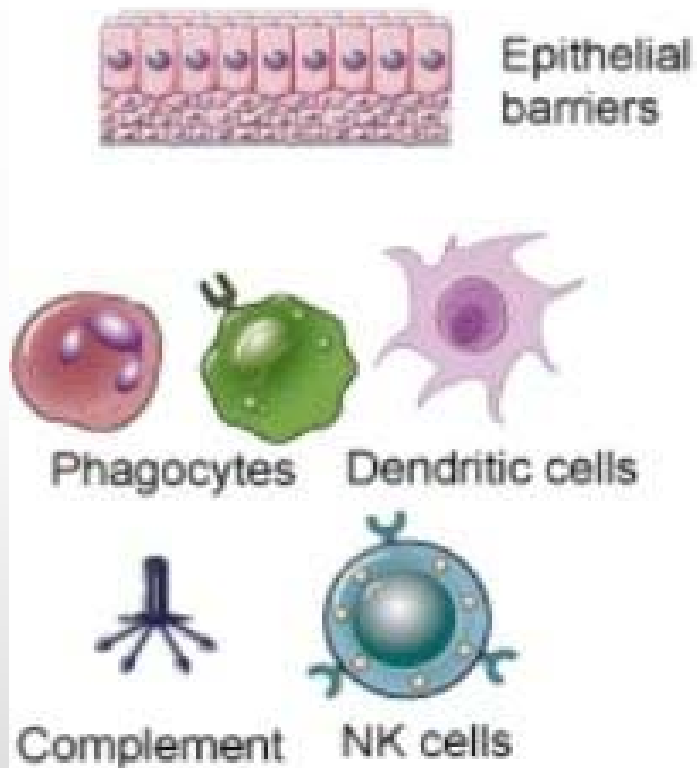
Innate immunity

- Innate immunity (also called natural, or native, immunity) is mediated by cells and proteins that are always present (hence the term innate), poised to react against infectious pathogens.
- *The major components of innate immunity are :*
 1. Epithelial barriers that block the entry of microbes .
 2. Phagocytic cells (mainly neutrophils and macrophages), dendritic cells (DCs), natural killer (NK) cells and other innate lymphoid cells .
 3. Several plasma proteins, including the proteins of the complement system .

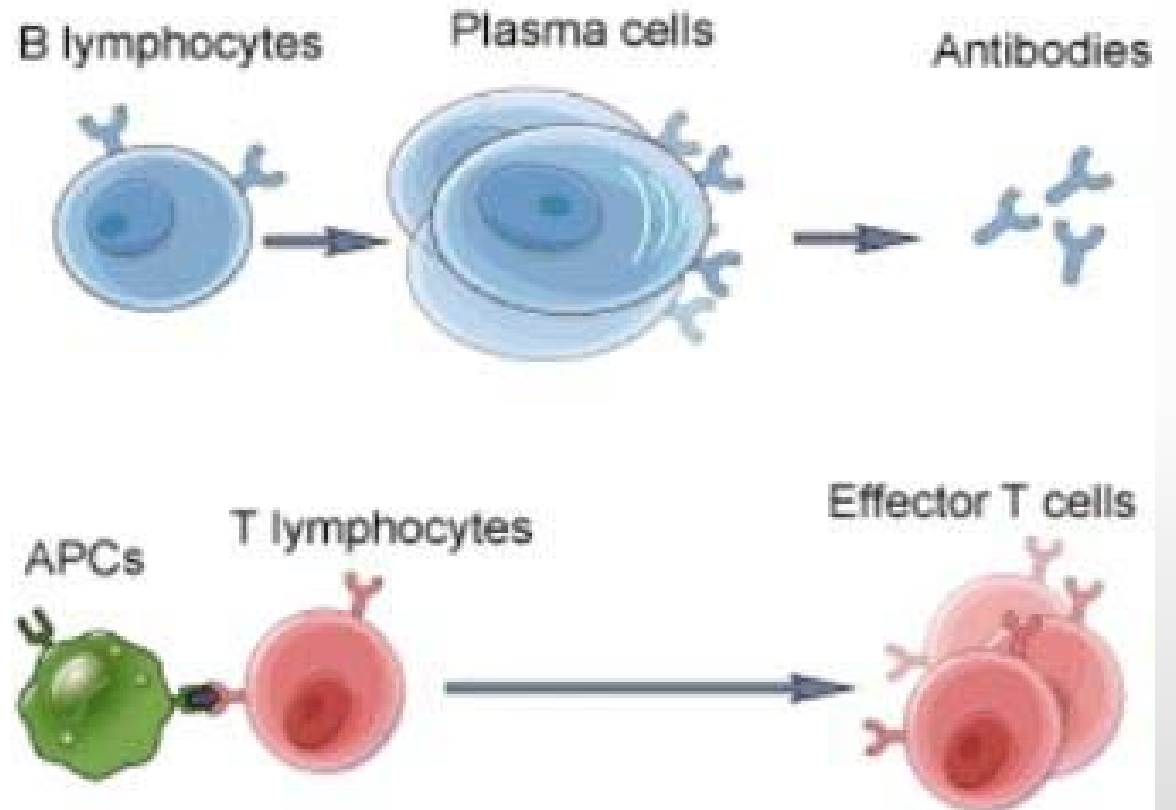
Adaptive immunity

- The adaptive immune system consists of lymphocytes and their products, including antibodies. In contrast to the limited repertoire of the innate immune system, the adaptive immune system can recognize a vast array of foreign substances.
- *There are two types of adaptive immunity :*
 1. **Humoral immunity**, mediated by soluble proteins called antibodies that are produced by B lymphocytes (also called B cells)
 2. **Cell-mediated (or cellular) immunity**, mediated by T lymphocytes (also called T cells).

Innate immunity



Adaptive immunity

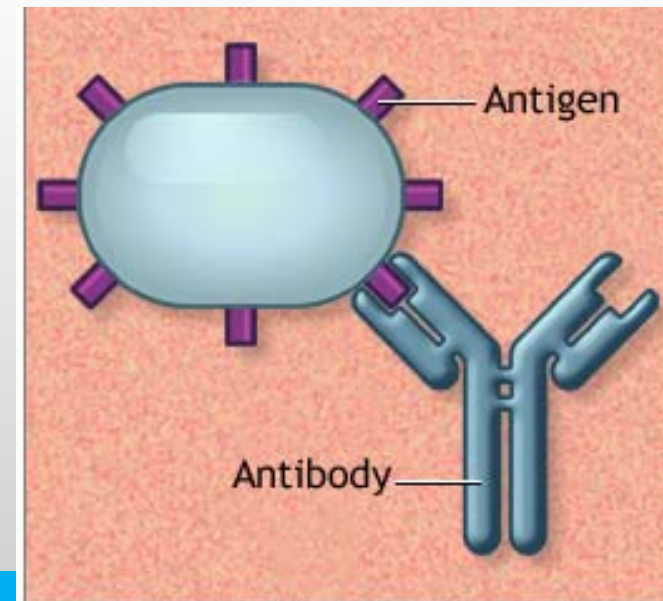


Important Definitions

- **Antigen** : is any substance , usually proteins,that can be recognized by the immune system and specifically bind to antibodies or to antigen receptors on lymphocytes .
- Antigen may be:
- Exogenous (e.g., microorganisms, toxins)
- Endogenous or self-antigens (as in autoimmune diseases)

Important Definitions

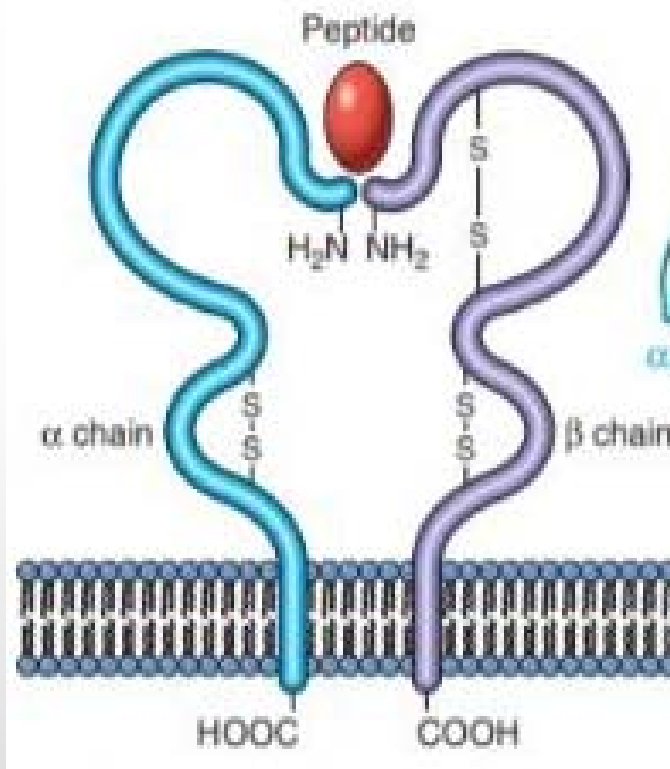
- **Antibody** : is an immunoglobulin produced by plasma cells derived from B lymphocytes, immunoglobulins= Ig = five classes: **IgG**, **IgM**, **IgA**, **IgE** and **IgD** , which specifically recognizes and binds to the antigen that induced its production, leading to:
 - Neutralization of toxins and viruses
 - Opsonization and enhanced phagocytosis
 - Activation of the complement system



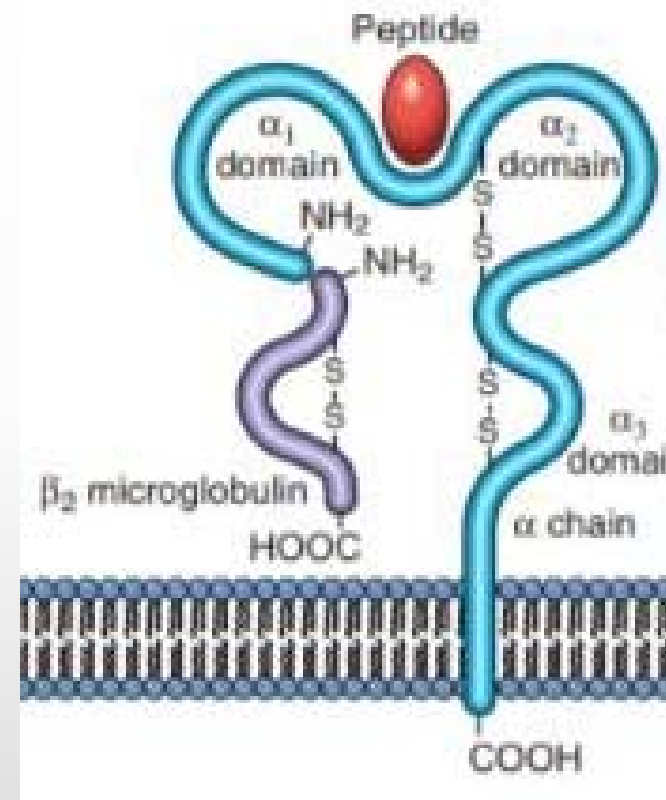
Important Definitions

Major Histocompatibility Complex (MHC) : is a group of highly polymorphic genes that encode cell surface proteins responsible for the presentation of peptide antigens to T lymphocytes, and thus play a central role in cell-mediated immune responses.

- There are two types of (MHC) :
 1. **MHC-I** which are found on all nucleated cells of the body.
 2. **MHC-II** which are found on macrophages, dendritic cells and B cells , antigen presenting cells (APCs).



MHC-II molecule

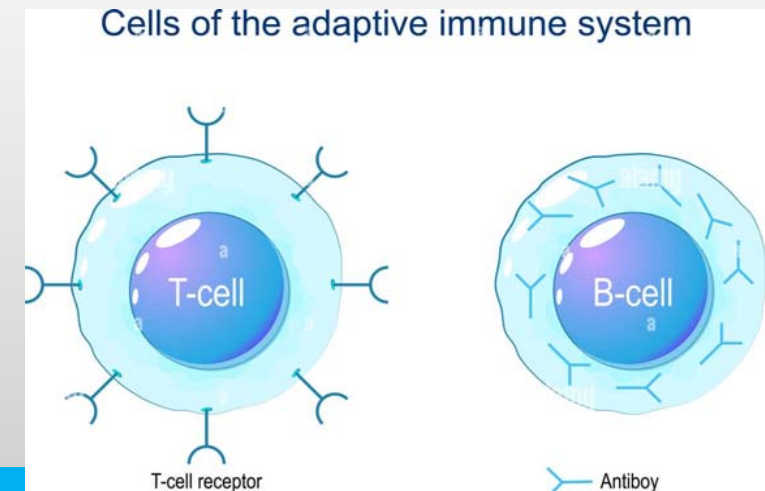


MHC-I molecule

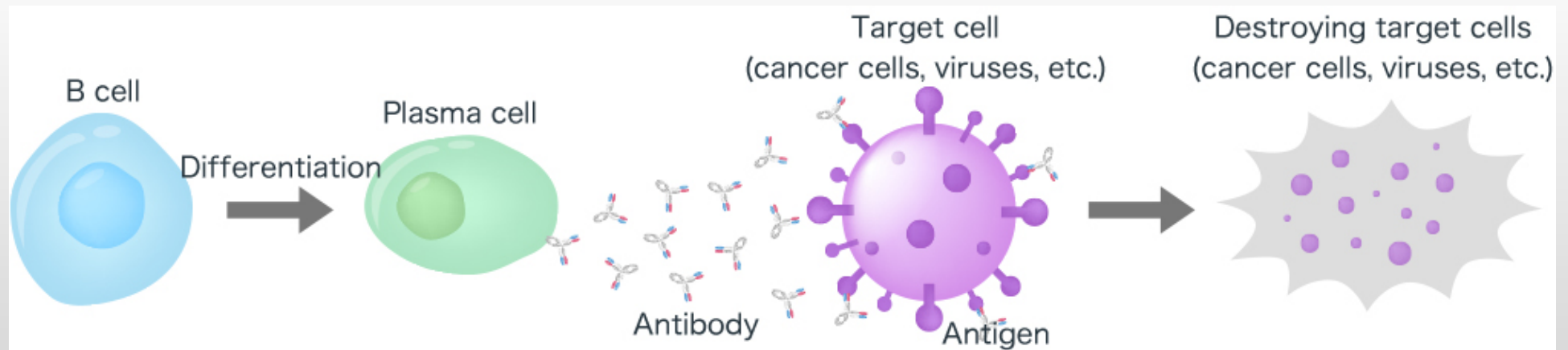
Cells of Immune System

1. **Lymphocytes** are the mediators of adaptive immunity and the only cells that produce specific and diverse receptors for antigens and can be classified into two types :

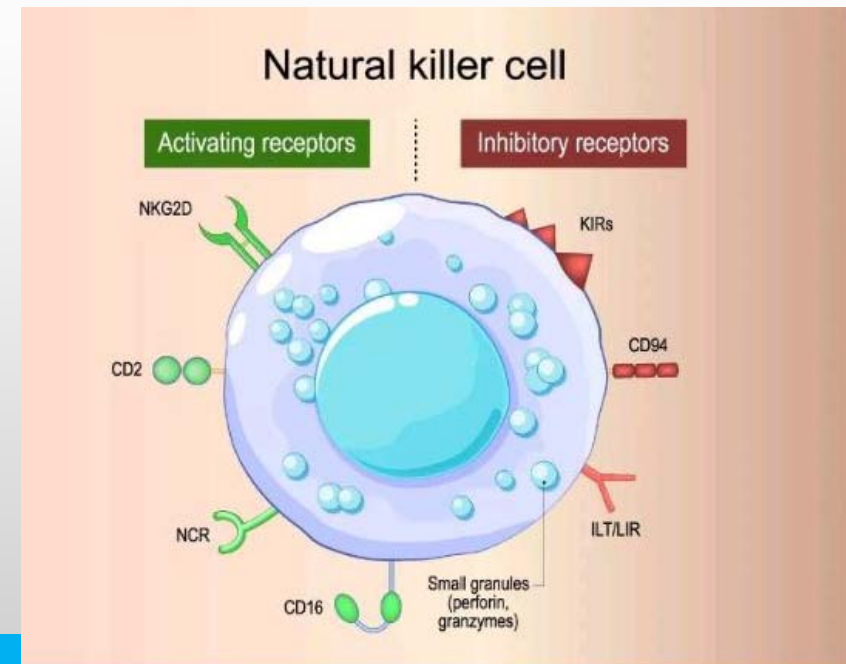
❖ **T (Thymus-derived) lymphocytes**: express antigen receptors called T cell receptors (TCRs) that recognize protein antigens that are displayed by MHC molecules on the surface of antigen-presenting cells.



- **B (Bone marrow-derived) lymphocytes:** B cells synthesize antibodies or immunoglobulins that recognize a wide variety of antigens. After stimulation, B cell will be differentiated into plasma cells which secrete large amounts of antibodies, which are the mediators of humoral immunity.



2. Natural killer (NK) cells kill cells that are infected by some microbes, or are stressed and damaged beyond repair. NK cells express inhibitory receptors that recognize MHC molecules that are normally expressed on healthy cells, and are thus prevented from killing normal cells.



3. Antigen-presenting cells (APCs) capture microbes and other antigens, transport them to lymphoid organs, and display them for recognition by lymphocytes. The most efficient APCs are:

- ❖ **Dendritic cells** which live in epithelia and most tissues .
- ❖ **Macrophages** which ingest microbes and other particulate antigens and display them for recognition by T lymphocytes.



Dendritic cell



Macrophage

- Under normal conditions, the immune response prevents disease, but occasionally, the inappropriate activation of the immune system can lead to debilitating or life-threatening illnesses, like:
 - 1- Allergic or hypersensitivity reactions.
 - 2- Autoimmune disorders.

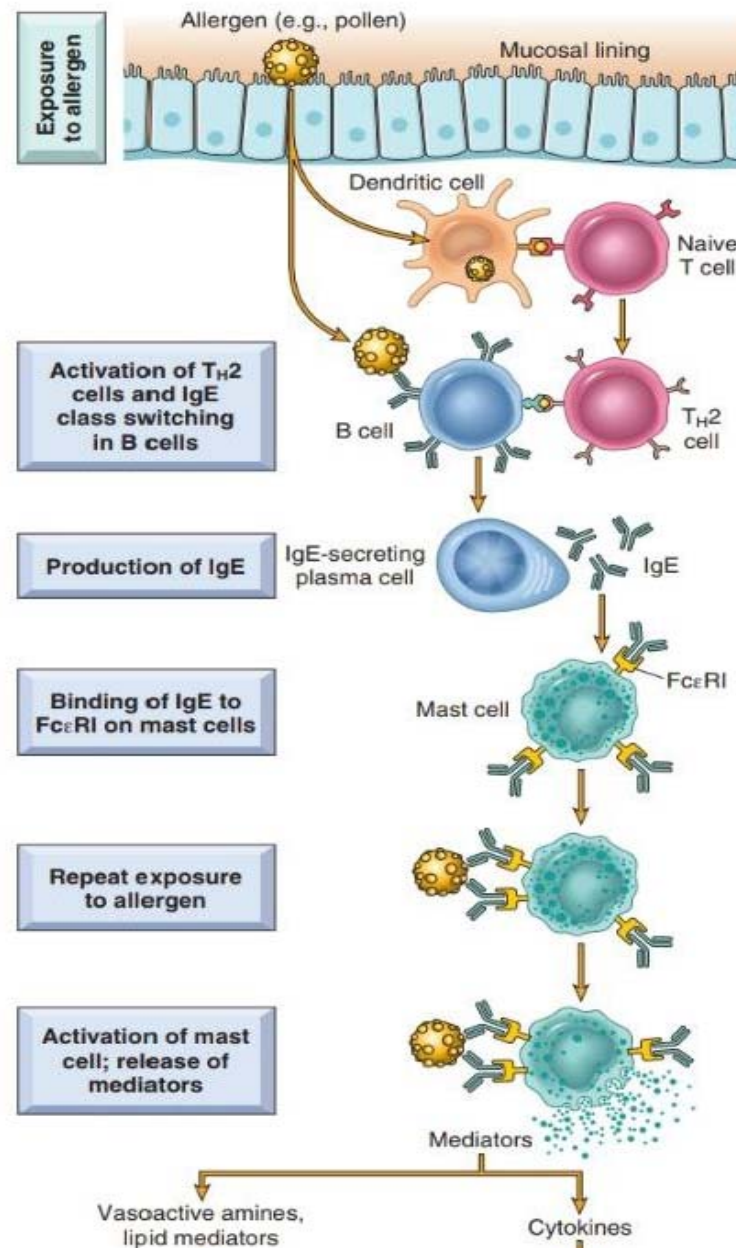
Allergic or hypersensitivity disorders

- **Hypersensitivity:** it is defined as an exaggerated immune response to a foreign agent resulting in injury to the host. It is caused by immune responses to environmental antigens called **allergens** that produce inflammation and cause tissue injury.
- **Allergens:** Any foreign substances capable of inducing an immune response. Exposure to the allergen can be through inhalation, ingestion, injection, or skin contact.

The main types of hypersensitivity reactions are as follows:

- Immediate (type I hypersensitivity)
- Antibody-mediated disorders (type II hypersensitivity)
- Immune complex–mediated disorders (type III hypersensitivity)
- T cell–mediated disorders (type IV hypersensitivity)

- **Immediate (type I hypersensitivity):** is also called an allergic reaction, induced by environmental antigens (allergens) that stimulate strong TH2 responses and IgE production in genetically susceptible individuals. IgE coats mast cells by binding to the receptor on the surface of mast cell .
- **Reexposure** to the allergen leads to activation of mast cells, and release of mediators. Principal mediators are histamine, proteases, and other granule contents that are responsible for the pathologic manifestations of immediate hypersensitivity.
- The clinical manifestations may be local or systemic, and range from mildly annoying rhinitis to fatal anaphylaxis.



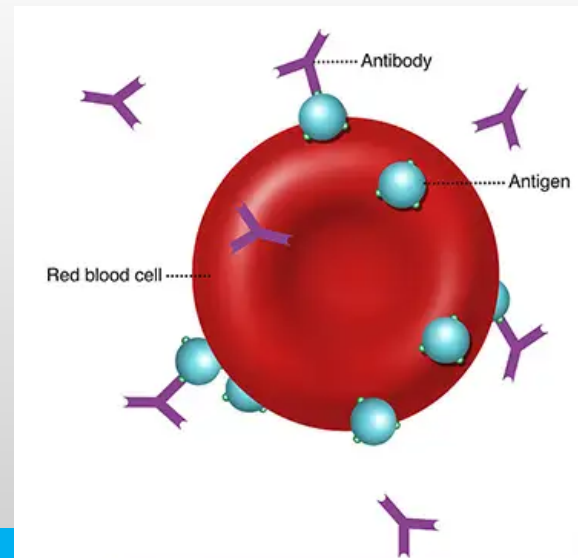


Rhinitis



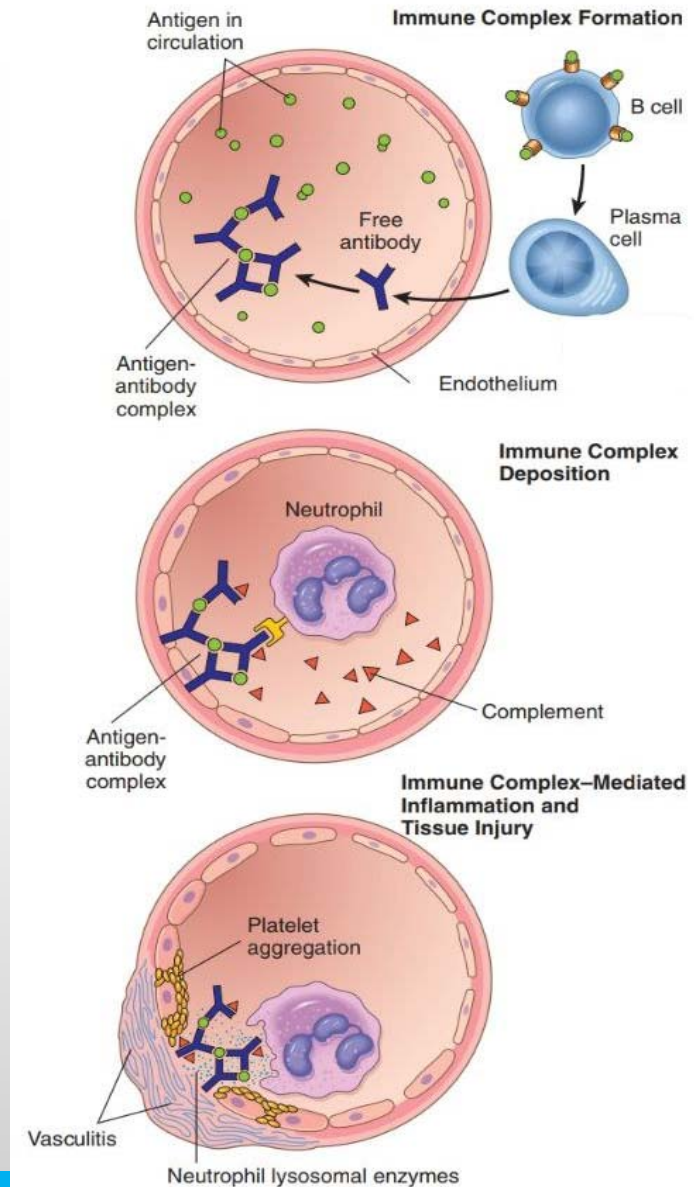
Anaphylaxis

- **Antibody-mediated disorders (type II hypersensitivity):** are caused by antibodies directed against target antigens on the surface of cells or other tissue components. The antigens may be normal molecules intrinsic to cell membranes or in the extracellular matrix, or they may be exogenous antigens (e.g., drug metabolite).
- These reactions are the cause of several important diseases .Examples of type II reactions include mismatched blood transfusions and certain drug reactions .



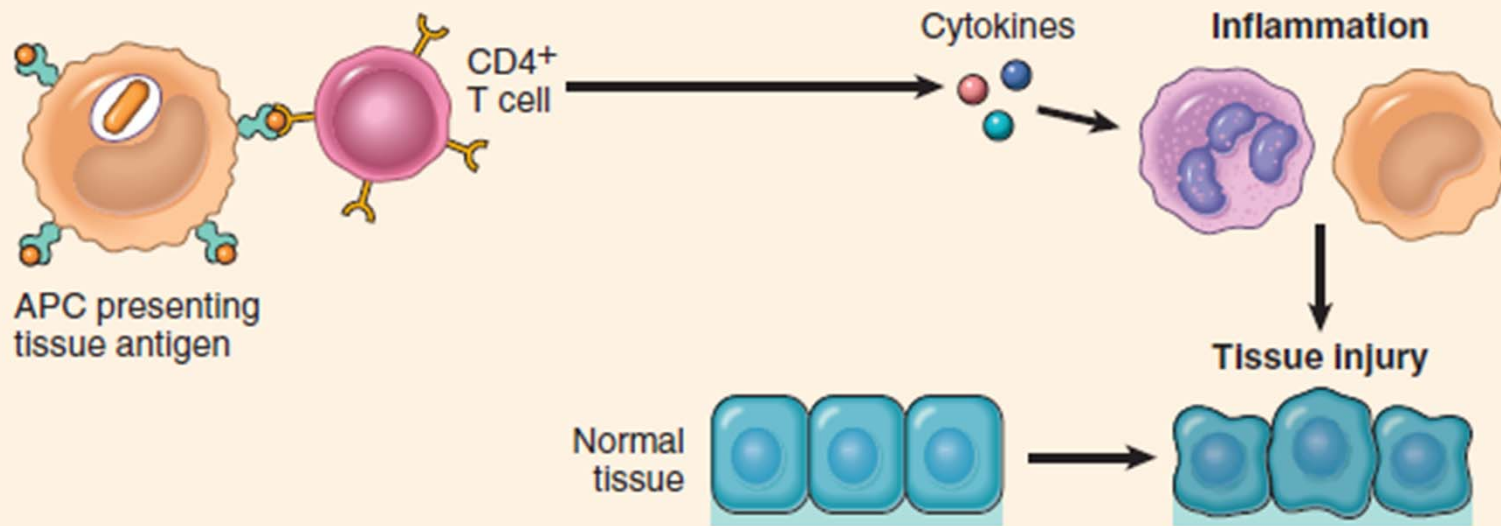
- **Immune complex-mediated disorders (type III hypersensitivity) :**

- Antigen-antibody (immune) complexes that are formed in the circulation may deposit in blood vessels or at extravascular sites where antigen may have been deposited that cause tissue damage by a variety of mechanisms .
- The most common sites of immune complex deposition are the kidney (glomerulonephritis), joints (arthritis), and small blood vessels (vasculitis).

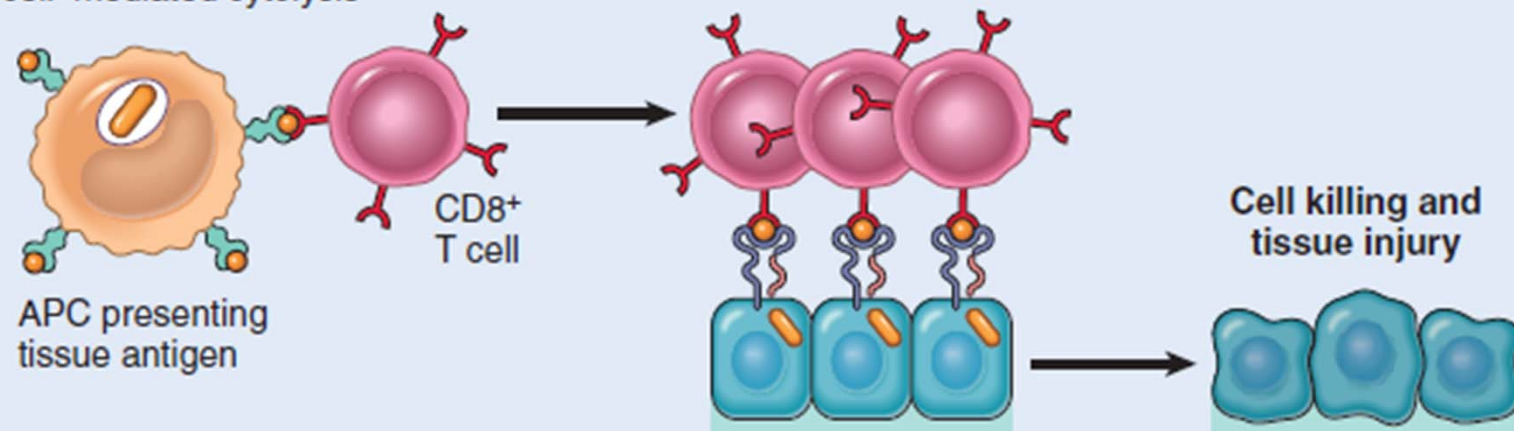


- **T cell-mediated disorders (type IV hypersensitivity)** : It is mediated by cells, not antibodies; occur 24 to 72 hours after exposure of a sensitized individual to the offending antigen. They are mediated by T lymphocytes that are directly cytotoxic or that secrete inflammatory mediators like cytokines that cause tissue changes.

Cytokine-mediated inflammation



T cell-mediated cytotoxicity



Autoimmune disorders

- Autoimmunity refers to immune reactions against self (“auto”) antigens. They are caused by a breakdown in the ability of the immune system to differentiate between self and nonself antigens. Normally, there is a high degree of immunologic tolerance to self-antigens, which prevents the immune system from destroying the host. Some autoimmune disorders are tissue specific; others affect multiple organs and systems.
- Examples : systemic lupus erythematosus and Sjögren syndrome

Thank You...